

SGH OUTPUT FINDINGS

SGH Prostate MRI Batch Inference Report

Overall Results

Metric	Count	%
Unique patient cases	3,538	
Successfully completed	3,473	98.2%
Failed	65	1.8%

One case (20201208_1debe8d041264ee1f4f8) was attempted twice and failed both times, hence 3,539 log entries for 3,538 unique patients.

Output Completeness (3,473 Successful Cases)

All 11 output artifacts are present at 100% for every successful case:

- T2, ADC, High-b NIfTI conversions
- Organ segmentation (NIfTI + RTSTRUCT)
- Lesion segmentation mask + 5 fold probability maps + merged probability
- Lesion RTSTRUCT
- Classifier report (lesions.txt)
- Combined organ+lesion RTSTRUCT

Each successful case produces an average of 184 output files.

Processing Performance

Metric	Value
Mean time per case	25.6 sec
Median time per case	24.0 sec
Fastest case	14.7 sec
Slowest case	79.2 sec

Metric	Value
Total compute time	~24.7 hrs

Processed over 2 days: 1,859 cases on Apr 29, 1,680 cases on Apr 30.

Case Distribution by Scan Year

Year	Cases
2020	2
2021	2
2023	975
2024	1,274
2025	1,284
Non-standard IDs	2

Failure Analysis (65 cases)

Category A - No output produced (47 cases)

DICOM-to-NIfTI conversion failed; all 3 modalities (T2, ADC, high-b) reported missing. Likely caused by incomplete or incompatible DICOM data at source. Includes 4 cases from 2020-2021 (very old scans).

Category B - Organ segmented but lesion inference failed (13 cases)

Organ segmentation completed successfully, but the lesion detection model crashed. Partial output (organ NIfTI + converted modalities) exists on disk.

Category C - Lesion mask produced but classifier report missing (4 cases)

Near-complete pipeline; the lesion probability maps and mask were generated, but the final classifier step (lesions.txt) failed. These cases have the most output artifacts among failures.

Category D - Non-standard patient IDs (2 cases)

IDs d6f57b6fef4174d2eaef24_113575d5e09041a0ac2e and efc4e68a3caf2273cbee26_c6e7ca742098fad2f0e0 do not follow the

YYYYMMDD_hash naming convention - likely test data or corrupted entries.

Key Takeaways

- 98.2% success rate across all 3,538 SGH cases
- 100% output completeness for all successful cases - every required artifact was generated.
- The 65 failures are well-characterized and recoverable: 47 are likely input data issues (missing/corrupt DICOM OR WRONG STUDY), 13 are lesion model failures, and 4 are minor classifier glitches.
- The 2 non-standard IDs can likely be excluded from the count, bringing the effective dataset to 3,536 real SGH cases with 3,473 (98.2%) successful.